

Offloading TC rules on OVS Internal Ports

John Hurley
presented by Simon Horman

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Internal Ports in OVS

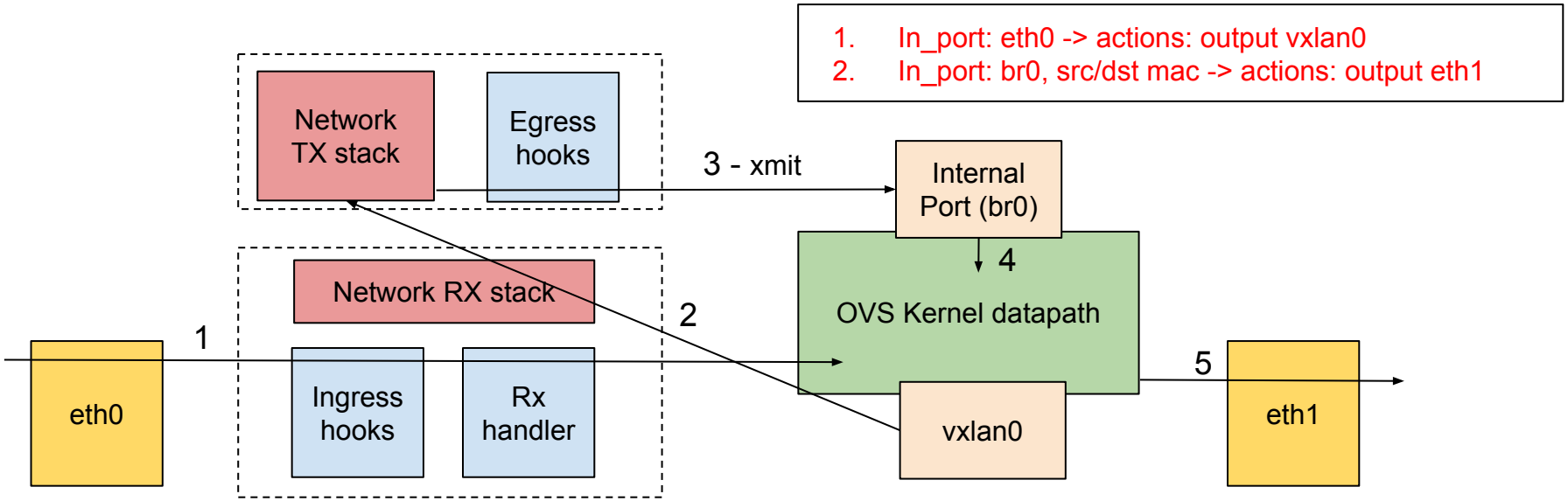
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- Registered in kernel by OVS and treated as standard netdevs
- Main differences with standard bridge devs:
  - If an OVS rule egresses to an internal port, the packet is passed to the network stack (ingress) rather than dev_queue_xmit
  - ndo_start_xmit() injects packet into OVS kernel datapath
- Rules applied to internal ports are not hit — TC ingress is bypassed
- Wish to address this to allow hardware offload of these rules

- TC rules outputting to an internal port should use Ingress redirect
- Two approaches to allowing TC to offload rules on internal ports:
  - Add TC ingress hook to OVS internal port modules
    - Required as it bypasses hook in the network stack
  - Offload rules as egress hooks
    - [RFC OVS 0/2] ovs-tc: support OVS internal port offload

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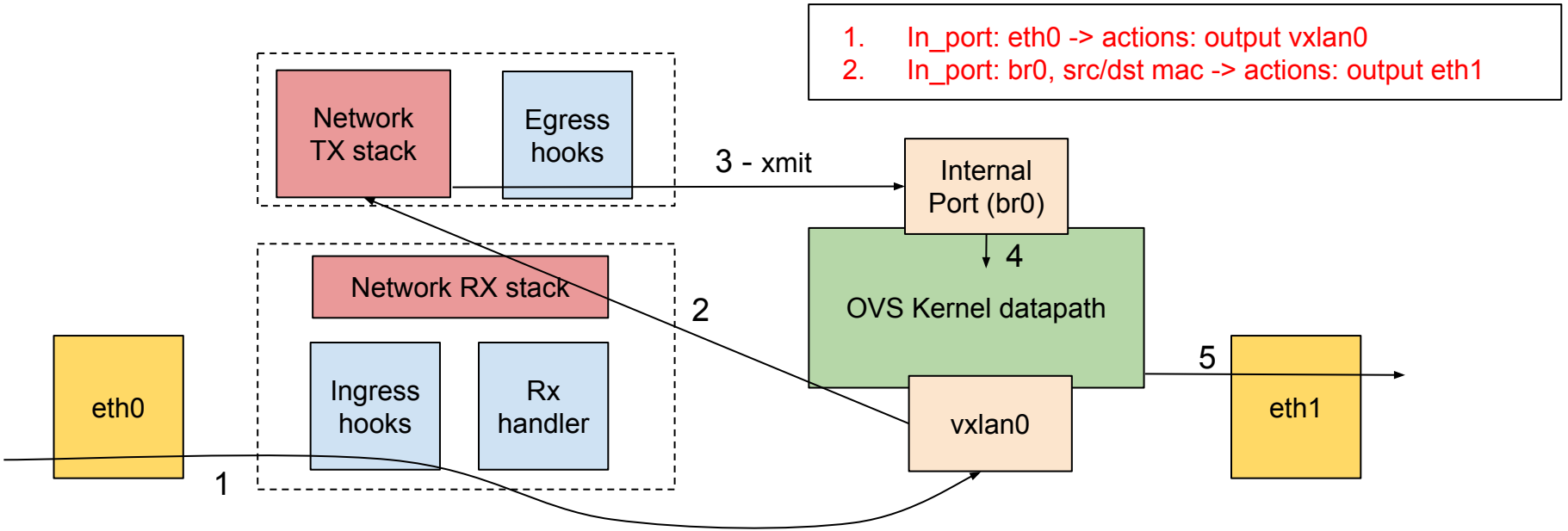
OVS Kernel (Encap with Tunnel IP on Bridge)



1. Packet enters OVS datapath and matches rule 1
2. Action is to egress port vxlan0
3. Passes ip_tun stack and finds br0 as next hop (neigh_output) and calls dev_queue_xmit on this
4. ndo_start_xmit on internal port pushes packet into OVS kernel datapath with br0 as ingress port
5. Matches rule 2 and egresses port eth1

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OVS-TC — Current

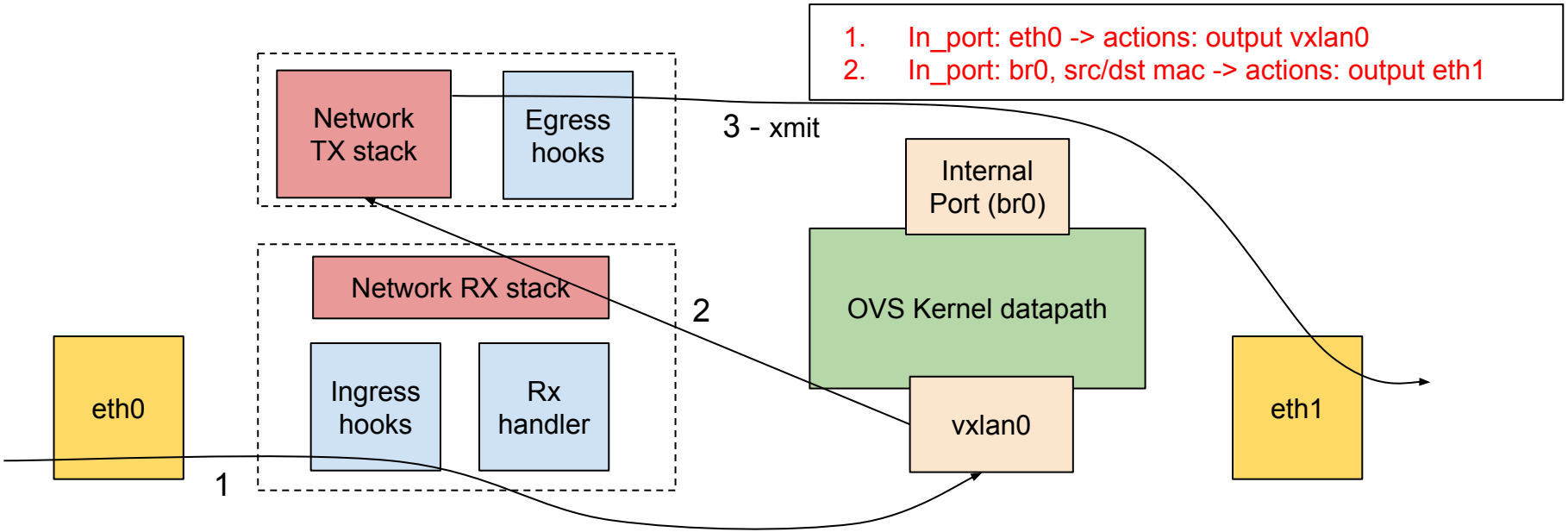


1. Rule 1 is offloaded as a TC flower ingress filter on eth0 so packet is directed straight to vxlan0 without entering the OVS datapath
2. Rule 2 is not offloaded to TC so path (2-5) through OVS kernel is taken

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OVS-TC — Modified to Use TC Egress Hook

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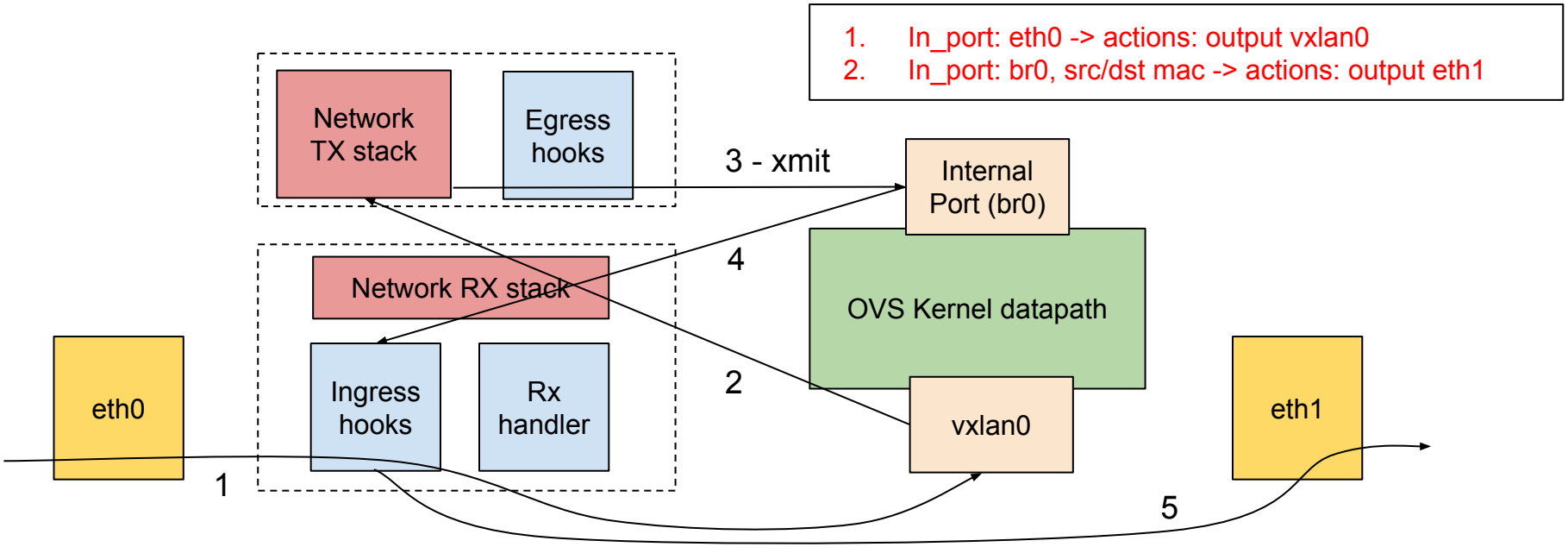


1. Packet matches ingress hook on eth0 and is directed to egress vxlan0
2. Next hop is determined as br0 so it is directed to egress that internal port
3. Rule 2 is installed on egress hook and packet is directed to eth1, bypassing OVS kernel datapath — egressing an internal port generates an OVS ingress action, so applying rule that match internal ports as egress hooks, achieves the same outcome as having an ingress hook in the internal port handlers

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OVS-TC — Modified to Use TC Ingress Hook

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1. Packet matches ingress hook on eth0 and is directed to egress vxlan0
2. Next hop is determined as br0 so it is directed to egress that internal port
3. When received by the internal port packet is passed to TC ingress hooks
4. Rule 2 is installed on ingress hook
5. Packet egresses via eth1

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